Kevin Chiaming **Chang**

PH.D. CANDIDATE | FORMAL VERIFICATION, CONTROL, MACHINE LEARNING

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Summary

UC Berkeley Ph.D. candidate developing formal-verification methods for learning-enabled control and hardware systems. Builds solver-backed algorithms that make neural controllers more interpretable and compositional, with research spanning exact controller transformation, contract-based safety analysis, optimization, and CPU microarchitecture verification.

Education

University of California, Berkeley — *Berkeley, CA*

Ph.D. in Electrical and Computer Engineering (GPA: 3.91/4.00) — *01/2021 – Present*

- Research focus: formal verification, learning-enabled cyber-physical systems, optimization, and contract-based design
- Advisor: Prof. Pierluigi Nuzzo; DesCyPhy Lab, Donald O. Pederson Center for Electronic Systems Design

National Taiwan University — *Taipei, Taiwan*

B.S. in Electrical Engineering (GPA: 3.67/4.00, Last 60: 4.00/4.00) — *09/2015 – 01/2020*

- Coursework in physical design, integrated circuits, computer architecture, machine learning, and deep learning
- Undergraduate research in electronic design automation and optimization

Research Experience

DesCyPhy Lab, Prof. Pierluigi Nuzzo — *Berkeley, CA*

Graduate Student Researcher — *01/2021 – Present*

Equivalent Neural-Controller Representations

- Developed a **provably exact transformation** from discrete-output ReLU neural controllers to soft decision trees, with automatic pruning of redundant branches.
- Demonstrated verification-time improvements of up to **20x** on MountainCar, CartPole, and CarRacing; resulted in first-author publications in **IEEE TAC (2026)** and **CDC (2023)**.

Compositional Safety for Learning-Enabled CPS

- Introduced robustness contracts that decouple plant dynamics from neural controllers and enable compositional closed-loop safety verification.
- Designed SMT- and MILP-based refinement algorithms to tighten abstraction bounds and synthesize realizable assume-guarantee specifications under bounded uncertainty.

Deep RL Controller Distillation for Verifiability

- Distilled deep-RL policies into compact decision-tree and kernel controllers, improving interpretability and validation while preserving task performance; resulted in an **Allerton 2022 publication**.

Electronic Design Automation Lab, Prof. Yao-Wen Chang — *Taipei, Taiwan*

Undergraduate Researcher — *09/2018 – 02/2020*

Routing and Optimization for EDA Contests

- Implemented a multithreaded detailed-routing engine that handled designs with up to **1M nets** using less than **64 GB** of memory in 10 hours on most benchmarks.
- Built routability-driven global- and system-level FPGA routers; contributed to **2nd place at ACM ISPD 2019**, a **Top 5 / Honorable Mention at ICCAD 2019**, and an **ICCAD 2021 publication**.

Institute of Information Science, Academia Sinica — *Taipei, Taiwan*

Software Researcher — *06/2018 – 09/2018*

- Implemented a semi-supervised domain-adaptation pipeline for semantic segmentation using unpaired image translation.
- Improved mean IoU on the Cityscapes benchmark by augmenting segmentation training with CycleGAN-generated domain information.

Skills

Focus	Formal verification, learning-enabled control, contract-based design, interpretable ML
Methods	SMT, MILP, assume-guarantee reasoning, symbolic abstraction, reachability analysis
Programming	Python, SystemVerilog, MATLAB, C++; PyTorch, scikit-learn
Tools	JasperGold, Xcelium, Genus, VCS, Gurobi, CVX
Related Areas	Semantic segmentation, domain adaptation, reinforcement learning, passivity, population games

Engineering Experience

Apple *Cupertino, CA*
Formal Verification Engineer Intern *05/2026 – 08/2026*

- Developed and debugged **formal properties** for CPU microarchitecture blocks, with emphasis on control-flow correctness.
- Diagnosed proof failures by refining assertions, assumptions, and coverage targets to distinguish design defects from underconstrained environments.

Teaching Experience

University of California, Berkeley *Berkeley, CA*
Graduate Student Instructor, Introduction to Design Automation *01/2026 – 05/2026*

— Developed end-to-end examples spanning RTL design, simulation, synthesis, timing analysis, and verification; supported students using industry EDA tools.

University of Southern California *Los Angeles, CA*
Graduate Student Instructor, Computer Architecture (3 semesters) *01/2022 – 05/2023*

— Developed examples and guided students through processor pipelines, out-of-order execution, hazards, caching, memory hierarchies, and performance tradeoffs.

Selected Technical Projects

Fantasy Football Performance Prediction *[Python/PyTorch/Scikit-Learn]*
Course Project, Pattern, Prediction, and Action *12/2025*

- Modeled weekly and season-long NFL player performance by combining box-score statistics, expert rankings, and qualitative commentary.
- Built a unified supervised-learning dataset and text embeddings to capture predictive signals beyond structured numerical features.

Sign Language Translation System *[C++/Python/PyTorch/Scikit-Learn]*
Arm Design Contest *09/2018*

- Built a real-time embedded sign-language translation system using a single 9-DOF IMU for gesture recognition.
- Integrated sensor processing and a learned gesture classifier in a live demonstration; won **2nd Place** and the **Best Popularity Award** among 130+ teams.

Selected Publications

1. Kevin Chang, Nathan Dahlin, Rahul Jain, and Pierluigi Nuzzo, "Equivalent and Compact Representations of Neural Network Controllers With Decision Trees," *IEEE Transactions on Automatic Control*, 2026, doi: 10.1109/TAC.2026.3676368.
2. Jair Certorio, Kevin Chang, Nuno C. Martins, Pierluigi Nuzzo, and Yasser Shoukry, "Passivity Tools for Hybrid Learning Rules in Large Populations," *Automatica*, vol. 187, Art. 112850, 2026.
3. Kevin Chang, Nathan Dahlin, Rahul Jain, and Pierluigi Nuzzo, "Exact and Cost-Effective Automated Transformation of Neural Network Controllers to Decision Tree Controllers," in *Proc. IEEE Conference on Decision and Control (CDC)*, pp. 7843–7848, 2023.
4. Nathan Dahlin, Kevin Chang, K. C. Kalagarla, Rahul Jain, and Pierluigi Nuzzo, "Practical Control Design for the Deep Learning Age: Distillation of Deep RL-Based Controllers," in *Proc. Allerton Conference on Communication, Control, and Computing*, 2022.
5. Wei-Kai Liu, Ming-Hung Chen, Chiaming Chang, Chen-Chia Chang, and Yao-Wen Chang, "Time-Division Multiplexing Based System-Level FPGA Routing," in *Proc. IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2021.

Selected Awards

Outstanding Performance Scholarship, *National Taiwan University* *Taipei, Taiwan*
Awarded for outstanding academic performance in electrical engineering *11/2019*

- Recognized for strong academic performance and undergraduate research contributions.

Top 5 and Honorable Mentions, *Cad Contest Problem C* *Westminster, USA*
LEF/DEF-based open-source global router *11/2019*

- Recognized in an ACM/IEEE contest field of 100+ teams.

2nd Place, *ACM ISPD Contest* *San Francisco, USA*
Initial detailed routing *04/2019*

- Placed 2nd among 33 international teams.

2nd Place and Best Popularity Award, *Arm Design Contest* *Taipei, Taiwan*
Sign language translation system *11/2018*

- Placed in the top 3 among 130+ hardware and embedded-systems teams.

Leadership

Chief Team Leader, *NTUEE Summer Orientation Camp* *Taipei, Taiwan*
11/2018

- Led a team of over 20 students to provide orientation information to 100+ undergraduates.

Vice Captain, *NTUEE Volleyball Team* *Taipei, Taiwan*
09/2017 – 06/2018

- Led two squads of 40+ players in a nationwide university competition with 50+ teams.